

Chapter 3

3.1. A continuous-time LTI system, when its input is $x(t) = u(t) - u(t-1)$, its zero state response is $y(t) = \delta(t+1) + \delta(t-1)$. The waveform of the signal $g(t)$ is shown in Figure 3-1, try to determine the zero state response $y_g(t)$ if input is $g(t)$. And sketch its waveform.

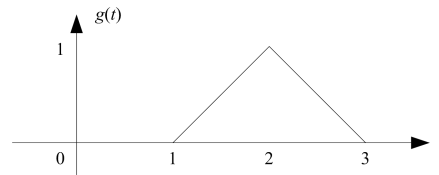


Figure.3-1

3.2. Consider a continuous-time LTI system with input $x(t) = u(t) - u(t-1)$ and output $y(t)$ shown in figure 3-2. When its input is unit step signal $u(t)$, try to determine zero state response and sketch its waveform.

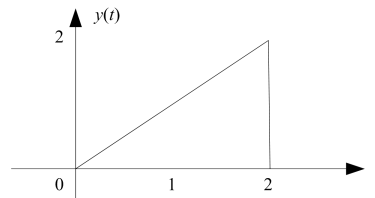


Figure.3-2

3.3. Consider a discrete-time LTI system with input $x_1[k]$ and zero state response $y_1[k]$. Try to determine zero state response $y_2[k]$ if input is $x_2[k]$. $x_1[k]$, $y_1[k]$ and $x_2[k]$ are shown in figure 3-3.

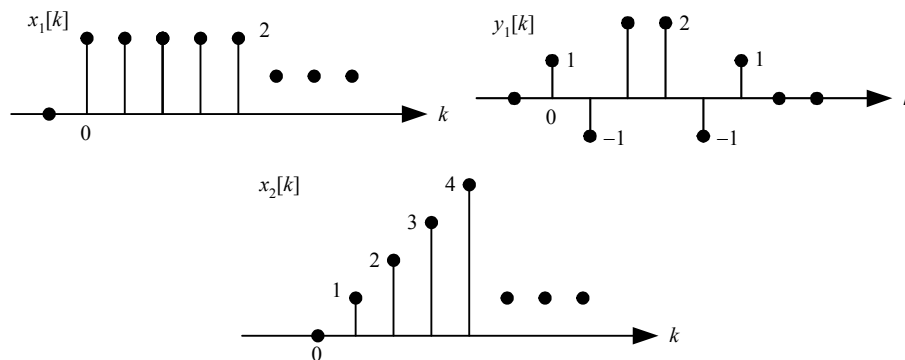


Figure.3-3

3.4. Try to determine the zero input response of the following continuous-time LTI system.

(1) $y''(t) + 5y'(t) + 4y(t) = 2x'(t) + 5x(t)$, $t > 0$; $y(0^-) = 1$, $y'(0^-) = 5$

(2) $y''(t) + 4y'(t) + 4y(t) = 3x'(t) + 2x(t)$, $t > 0$; $y(0^-) = -2$, $y'(0^-) = 3$

(3) $y''(t) + 4y'(t) + 8y(t) = 3x'(t) + x(t)$, $t > 0$; $y(0^-) = 5$, $y'(0^-) = 2$

(4) $y'''(t) + 3y''(t) + 2y'(t) = x'(t) + 4x(t)$, $t > 0$; $y(0^-) = 1$, $y'(0^-) = 0$, $y''(0^-) = 1$

3.5. Try to determine the impulse response $h(t)$ of continuous-time LTI system described by the following differential equations.

- (1) $y'(t) + 3y(t) = 2x(t)$, $t > 0$;
- (2) $y'(t) + 6y(t) = 3x'(t) + 2x(t)$, $t > 0$
- (3) $y''(t) + 3y'(t) + 2y(t) = 4x(t)$, $t > 0$;
- (4) $y''(t) + 4y'(t) + 4y(t) = 2x'(t) + 5x(t)$, $t > 0$;
- (5) $y''(t) + 4y'(t) + 8y(t) = 4x'(t) + 2x(t)$, $t > 0$;
- (6) $y''(t) + 5y'(t) + 6y(t) = x''(t) + 7x'(t) + 4x(t)$, $t > 0$;

3.6. Try to determine impulse response and step response of inductor current of RL circuits shown in figure 3-4.

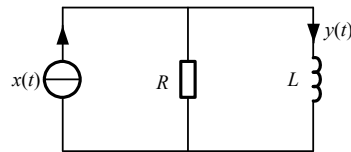


Figure.3-4

3.7. Calculated convolution integral of signal $x(t)$ and $h(t)$ shown in figure 3-5 by graphical method.

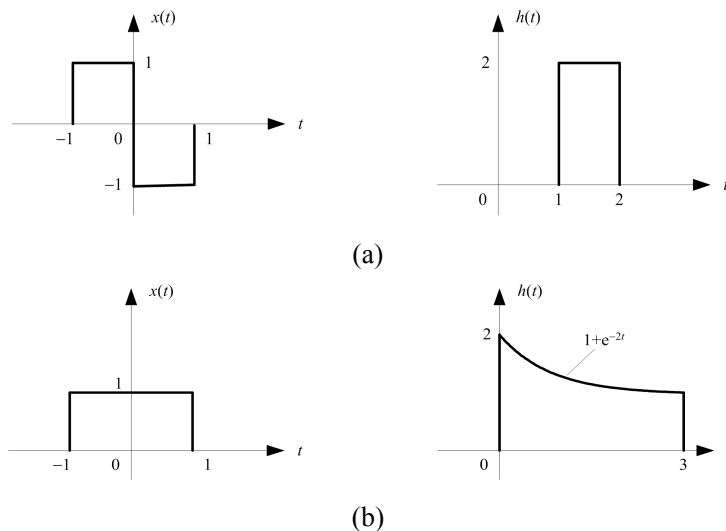
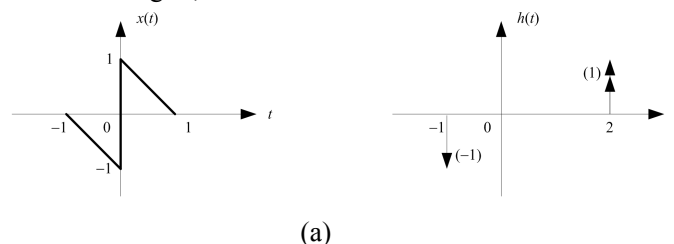


Figure.3-5

3.8. Calculating the following convolution integral.

- (1) $[\delta(t+1) + 2\delta(t-1)] * [\delta(t-1) - \delta(t-3)]$
- (2) $[u(t) - u(t-1)] * [u(t-2) - u(t-3)]$
- (3) $[u(t) - u(t-1)] * e^{-2t}u(t)$
- (4) $2e^{-2t}u(t) * 3e^{-t}u(t)$
- (5) $2e^{-2(t-1)}u(t-1) * 3e^{-t}u(t-2)$
- (6) $2e^{-2t}u(t) * 3e^t u(-t)$
- (7) $2e^{2t}u(-t) * 3e^t u(-t)$,
- (8) $2e^{-2t}u(t) * 3e^{-|t|}$

3.9. Calculate the convolution integral of signal $x(t)$ and $h(t)$ shown in figure 3-6 by using the properties of convolution integral, and sketch the result waveform.





(b)

Figure.3-6

- 3.10. We know $x(t)$ shown in figure 3-7(a) is a triangular wave, $\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$ shown in figure 3-7(b) is unit impulse sequence, try to calculate $x(t) * \delta_T(t)$ and sketch convolution result waveform. Where $T = 3\tau$, $T = 2\tau$ and $T = \tau$.

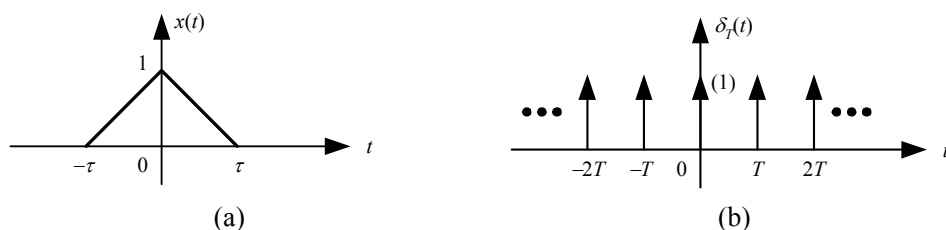


Figure.3-7

- 3.11. Consider a continuous-time LTI system with input $x(t)$ and step response $g(t)$. Try to prove the zero state response of the system is

$$y_{zs}(t) = \int_{-\infty}^{\infty} x'(\tau)g(t - \tau)d\tau$$

The formula is called Duhamel expression.

- 3.12. We know the differential equation of a causal LTI system is

$$y''(t) + 7y'(t) + 10y(t) = 2x'(t) + 3x(t), \quad t > 0$$

And $x(t) = e^{-t}u(t)$, $y(0^-) = 1$, $y'(0^-) = 1$, try to calculate:

- (1) Unit impulse response $h(t)$ of the system.
- (2) Zero input response $y_{zi}(t)$, zero state response $y_{zs}(t)$ and complete response $y_1(t)$ of system.
- (3) If input is $x(t) = e^{-t}u(t - 1)$, try to calculate complete response $y_2(t)$.

- 3.13. Try to find the zero input response of the following continuous-time LTI system.

$$(1) \quad y[k] - \frac{5}{2}y[k-1] + y[k-2] = x[k] + x[k-1], \quad k \geq 0, \quad y[-1] = 1, \quad y[-2] = 3$$

$$(2) \quad y[k] - y[k-1] + y[k-2] = x[k], \quad k \geq 0, \quad y[-1] = 0, \quad y[-2] = 1$$

$$(3) \quad y[k] + 2y[k-1] + y[k-2] = 2x[k] + x[k-1], \quad k \geq 0, \quad y[-1] = -1, \quad y[-2] = 3/2$$

- 3.14. Try to find the unit impulse response $h[k]$ of discrete-time LTI system described by the following equations.

$$(1) \quad y[k] = x[k] - x[k-1]$$

$$(2) \quad y[k] + 0.6y[k-1] = x[k]$$

$$(3) \quad y[k] + 0.6y[k-1] = 2x[k-1]$$

$$(4) \quad y[k] - \frac{3}{4}y[k-1] + \frac{1}{8}y[k-2] = x[k]$$

$$(5) \quad y[k] - 5y[k-1] + 6y[k-2] = x[k] - 3x[k-2]$$

3.15. Proving: if range without zero of $x[k]$ is $N_1 \leq k \leq N_2$, and range without zero of $h[k]$ is $N_3 \leq k \leq N_4$, we can find that range without zero of $x[k] * h[k]$ is $N_1 + N_3 \leq k \leq N_2 + N_4$.

3.16. Calculate $x[k] * h[k]$:

- (1) $x[k] = \{1, 2, 1\}$, $h[k] = \{1, 0, 2, 0, 1\}$
- (2) $x[k] = \{-3, 4, 6, 0, -1\}$, $h[k] = u[k] - u[k - 4]$
- (3) $x[k] = \left(\frac{1}{2}\right)^k u[k - 2]$, $h[k] = \delta[k] - \delta[k - 1]$
- (4) $x[k] = \delta[k + 1] + \delta[k - 1]$, $h[k] = \sum_{r=-\infty}^{\infty} \delta[k - 4r]$

3.17. We know the difference equation of a discrete-time LTI system is

$$y[k] - \frac{5}{6}y[k - 1] + \frac{1}{6}y[k - 2] = x[k]$$

$y[-1] = 0$, $y[-2] = 1$, $x[k] = u[k]$, Try to calculate:

- (1) Unit impulse response of the system $h[k]$
 - (2) Zero input response $y_{zi}[k]$, zero state response $y_{zs}[k]$ and complete response $y[k]$ of system.
 - (3) Transient response component $y_t[k]$, steady-state response component $y_s[k]$, inherent response component $y_h[k]$ and forced response component $y_p[k]$.
- 3.18. Try to find the unit impulse response $h(t)$ of continuous-time LTI system shown in figure 3-8. And $h_1(t) = u(t)$, $h_2(t) = 2\delta(t - 2)$, $h_3(t) = e^{-2t}u(t)$, $h_4(t) = e^{-3t}u(t)$.

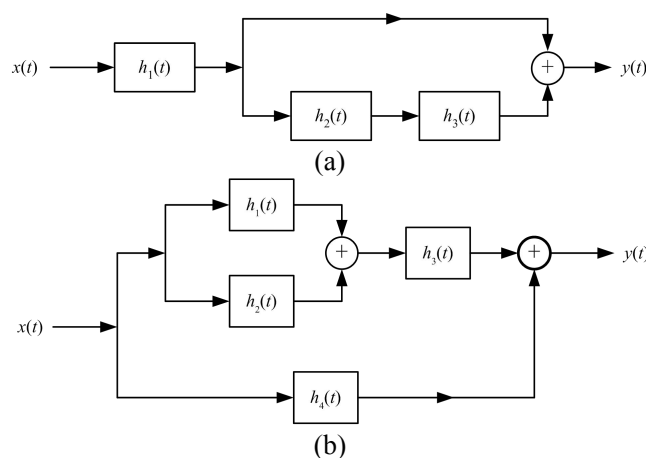


Figure 3-8

3.19. Determine whether the following LTI systems are causal system and stable system.

- (1) $h(t) = e^{-|t|} \sin(\omega_0 t)$
- (2) $h(t) = e^{at} u(-t)$
- (3) $h[k] = \cos\left(\frac{\pi}{2}k\right)u[k]$
- (4) $h[k] = a^k \{u[k] - u[k - N]\}$ ($N > 0$)